

DATA.md — what is in this repo, what is not, and how to get the rest

This project uses five datasets. **None of the raw third-party data is redistributed here** — it is external data under its own licence, and this repo pulls it straight from the primary sources with `scripts/fetch_*.py`. What is committed is (a) our own derived artifacts, small enough to live in git, and (b) a precomputed results store shipped as a GitHub Release asset.

The result: you can clone the repo and **run the tests and the offline dashboard immediately**, without downloading a single byte of raw NWP data.

1. What is committed directly in the repo (~42 MB)

These are **our derived products**, not third-party data. Safe to redistribute, small enough for git, and they make the repo runnable.

Path	Size	What it is
<code>data/processed/dataset.parquet</code>	34 MB	The full labelled modelling frame: 279,650 rows × 93 columns (subdivision × init × lead, with the bust label, all 52 features, and the split tag). This is the single file <code>train_model.py</code> needs.
<code>data/interim/imd_subdivisions.gpkg</code>	6 MB	The 34 modelled IMD subdivision polygons (built as unions of 641 Census-2011 districts). The API serves <code>/api/regions</code> from this.
<code>data/interim/truth_subdivision_daily.parquet</code>	180 KB	Observed subdivision-daily rainfall (area-mean), derived from the IMD gridded product.
<code>data/interim/weights_*.parquet</code>	~100 KB	Cached grid↔polygon area-overlap weights, keyed by grid geometry.
<code>data/artifacts/bust_model.joblib</code>	1.5 MB	The trained XGBoost booster + isotonic calibrator + feature list.
<code>data/artifacts/results.json</code>	7.5 KB	Held-out-year evaluation table (all baselines + model, per-lead, reliability).
<code>data/artifacts/*.csv</code>	<5 KB each	Ablation, SHAP importance, stress-test, and ENS-baseline comparison tables.
<code>data/artifacts/reference_distributions.npz</code>	134 KB	Training-year feature distributions, used by the drift monitor.

Derived-data provenance note: `dataset.parquet` and `truth_subdivision_daily.parquet` contain an observed-rainfall column derived from IMD's 0.25° gridded product (area-averaged to subdivisions, reformatted, and joined with model outputs). This is a transformative derivative work under IMD's research-use terms; attribution is in `LOGIC.md` §5. It is not a copy of the IMD product and cannot be used to reconstruct it.

2. What ships as a GitHub Release asset (~60 MB)

Asset	Size	What it is
<code>bulletins.sqlite</code>	63 MB	The precomputed bulletin store: 79,900 pre-scored (subdivision × init × lead) rows, each with the calibrated bust probability, bagged prediction interval, OOD status, SHAP reason strings, regime vector, and the baseline probability. This is what the offline dashboard serves.

It is a Release asset rather than a committed file because 60 MB of binary would bloat every clone of the git history. It is **fully regenerable** from the committed `dataset.parquet` + `bust_model.joblib` by running `scripts/generate_bulletins.py` (~100 s), so the Release is a convenience, not a dependency.

Get it (either works):

```
# One command – downloads and verifies the checksum:
PYTHONPATH=src python scripts/fetch_release_artifacts.py

# ...or regenerate it locally from the committed artifacts:
PYTHONPATH=src python scripts/generate_bulletins.py
```

SHA-256 of `bulletins.sqlite` (v0.1.0): `5cbb61350c2ba63dbcf2dc00bc1d13b9a3d2f9a70e66d8317121cf996e6f241c`

3. What is NOT in the repo and never will be (~570 MB of raw data)

This is external third-party data. We do not rehost it — we cite it and reproduce it from the primary source. Rehosting it would be a licensing grey area and would teach the wrong lesson about how research artifacts are shared.

Dataset	Local size	Primary source	Fetch with
IMD 0.25° gridded rainfall (2016–2022)	170 MB	<code>imdpune.gov.in/cmpg/Griddata/Rainfall_25_NetCDF.html</code>	<code>scripts/fetch_imd.py</code>
WeatherBench 2 IFS HRES forecasts	72 MB (of ~9 GB transferred)	<code>gs://weatherbench2/datasets/hres/...</code> (anonymous GCS)	<code>scripts/fetch_hres.py</code>
WeatherBench 2 ERA5 analysis	~450 MB	<code>gs://weatherbench2/datasets/era5/...</code>	<code>scripts/fetch_era5.py</code>
WeatherBench 2 IFS ENS (50-member)	360 KB (subsampled)	<code>gs://weatherbench2/datasets/ifs_ens/...</code>	<code>scripts/fetch_ens.py</code>
Census-2011 district boundaries	10 MB	<code>github.com/datameet/maps</code>	<code>scripts/fetch_boundaries.py</code>

Full reproduction from nothing:

```
python scripts/fetch_imd.py
python scripts/fetch_hres.py
python scripts/fetch_era5.py
python scripts/fetch_ens.py --year 2019 --every 6
python scripts/fetch_ens.py --year 2020 --every 6
python scripts/fetch_ens.py --year 2022 --every 3
python scripts/fetch_boundaries.py
# then rebuild everything:
python scripts/build_truth.py
python scripts/build_dataset.py
python scripts/train_model.py
python scripts/generate_bulletins.py
```

Total download time is dominated by the WeatherBench 2 HRES pull (~20 min) because WB2's zarr chunks span the globe (you download the world and slice to India locally — see [DECISIONS.md](#) D-002).

4. The three tiers, at a glance

└ committed in git (~42 MB) ─────────┐ our derived products: dataset.parquet, trained model, geometry, eval tables	clone the repo -> tests pass, model loads, pipeline runs
└ GitHub Release asset (~60 MB) ─────────┐ bulletins.sqlite (precomputed dashboard)	one command -> offline dashboard runs
└ never in the repo (~570 MB, external) ─────────┐ raw IMD / WeatherBench 2 / Census data	scripts/fetch_*.py -> reproduce from source

This is the standard pattern for a reproducible-research repo: **cite the primary source, ship the reproduction script, host only your own derivatives**. It is what makes the repo credible rather than merely convenient.